

Answer Key — Biology Grade 11

Sub-Strand 1.2: Ecology • Lesson 2: Who Lives Where? Habitat Classification and Adaptations in a Kenyan River and Beyond

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	C	A	D	D	B	B

USE AFTER: PREDICT PHASE — Predict: six quick prompts on the board (mangrove at Mombasa, Mida Creek)

Q1. Mangrove trees grow in the salty mud at Mida Creek near Mombasa. Which habitat group do mangroves belong to?

- A. Xerophytes
- B. Mesophytes
- C. Halophytes**
- D. Hydrophytes

Correct answer: C. Mangroves live in salty conditions, so they are halophytes. Learners may say hydrophyte because mangroves stand in water, but the key habitat condition is salt.

USE AFTER: OBSERVE PHASE — Jigsaw Round 1: expert group on xerophytes (Euphorbia, sisal, Opuntia, acacia)

Q2. Euphorbia in dry Turkana has a thick, fleshy stem covered with a waxy layer. How does this waxy layer help it survive?

- A. It removes extra salt from the plant
- B. It helps the plant take in dissolved oxygen
- C. It reduces the amount of water lost from the plant**
- D. It helps the plant float on water

Correct answer: C. Euphorbia is a xerophyte. A thick waxy cuticle reduces water loss, which is vital in a dry habitat.

USE AFTER: OBSERVE PHASE — Jigsaw Round 1: expert group on hydrophytes (water hyacinth, water lily, papyrus)

Q3. Water hyacinth on Lake Victoria has leaves with large air spaces inside. What is the main use of these air spaces?

- A. They help the leaves float at the water surface**
- B. They store water for the dry season
- C. They let the plant lose water quickly in hot weather
- D. They let the plant breathe through gills

Correct answer: A. Water hyacinth is a hydrophyte. Air spaces in its leaves and stems make it buoyant, so it floats and gets light at the surface.

USE AFTER: EXPLAIN PHASE — Jigsaw Round 2: oral question 'Why can a tilapia not live in a desert?'

Q4. Why can a tilapia not live in a desert?

- A. It cannot grow thick fur for the cold desert nights
- B. It has a hump that stores too much water
- C. Desert air has too much oxygen for it
- D. Its gills only work in water, where they take in dissolved oxygen**

Correct answer: D. Aquatic animals such as tilapia use gills to get dissolved oxygen from water. Without water they cannot get oxygen, so they cannot survive in a desert.

USE AFTER: EXPLAIN PHASE — Jigsaw Round 2: claim-evidence-reasoning on why algae and aquatic weeds grow well while fish decline

Q5. A Kenyan river once had many fish, but now has much algae and many weeds and sometimes smells bad. Which explanation best fits what the class learned about habitats and adaptations?

- A. Algae and weeds are xerophytes, so they grow best when the river dries up
- B. The fish are old and the algae grow because the fish have stopped eating them
- C. The fish have no food, so the algae and weeds have taken over the river
- D. Nutrient-rich, warm water favours algae and weeds. When they die and decay, they use up dissolved oxygen, which harms fish that breathe with gills**

Correct answer: D. Hydrophytes and algae thrive in nutrient-rich water. Their decay lowers dissolved oxygen and causes smell, and fish depend on that oxygen. The other options repeat the learners' first ideas or misname the plants.

USE AFTER: EXPLAIN PHASE — Jigsaw Round 2: teacher addresses the misconception about how adaptations arise (camel, desert animals)

Q6. A camel in Marsabit has a hump and can go for long periods without drinking. How did camels come to have these adaptations?

- A. The desert heat changes the body of every young camel into a desert type
- B. Camels with helpful features survived and passed them on over many generations**
- C. Camels chose to develop a hump when they entered the desert
- D. Each camel grew a hump during its life because it needed water

Correct answer: B. Adaptations are not chosen by the organism. They arise over generations through natural selection, as individuals with useful features survive and reproduce.

USE AFTER: MODEL BUILDING PHASE — Model: revising the river ecosystem sketch with the example arrow 'Nutrients → algae → less oxygen → fish?'

Q7. In a model of the river, a learner draws the arrow 'Nutrients → algae → less oxygen → fish?' and puts a question mark at the end. Why is the question mark used?

- A. Because fish do not need oxygen
- B. Because this is only a possible link until there is evidence to support it**
- C. Because the arrows must always end with a question mark
- D. Because algae are not found in rivers

Correct answer: B. The arrow is a possible link between nutrients, algae, oxygen and fish. The question mark shows it is still an idea to be tested with evidence in later lessons.